

DPEN022: Enabling Mathematics B

Class Questions Week 6A

Some questions only include the final answer and do not have fully worked solutions.

Week 6A

Key Idea – Simple Substitution with other Functions

1. (a) $\int \cos x \sqrt{\sin x - 1} \, dx$

Step 2: $\frac{du}{dx} = \cos x$ and $du = \cos x \, dx$

Step 3: $\int \cos x \sqrt{\sin x - 1} \, dx = \int \sqrt{\sin x - 1} \cos x \, dx$
 $= \int \sqrt{u} \, du$
 $= \int u^{\frac{1}{2}} \, du$

Step 4: $= \frac{u^{\frac{3}{2}}}{\frac{3}{2}} + C$
 $= \frac{2u^{\frac{3}{2}}}{3} + C$

Step 5: $= \frac{2(\sin x - 1)^{\frac{3}{2}}}{3} + C = \frac{2\sqrt{(\sin x - 1)^3}}{3} + C$

(b) $\int \frac{e^{2x}}{(e^{2x} - 3)^7} \, dx$

Step 1: Let $u = e^{2x} - 3$

Step 2: $\frac{du}{dx} = 2e^{2x}$ and $du = 2e^{2x} \, dx \Rightarrow \frac{du}{2} = e^{2x} \, dx$

Step 3: $\int \frac{e^{2x}}{(e^{2x} - 3)^7} \, dx = \int \frac{1}{u^7} \times \frac{du}{2}$
 $= \frac{1}{2} \int \frac{1}{u^7} \, du = \frac{1}{2} \int u^{-7} \, du$

Step 4: $= \frac{1}{2} \times \frac{u^{-6}}{-6} + C$
 $= -\frac{u^{-6}}{12} + C$

Step 5: $= -\frac{(e^{2x} - 3)^{-6}}{12} + C = -\frac{1}{12(e^{2x} - 3)^6} + C$

(c) $\int e^{\cos x} \sin x \, dx = -e^{\cos x} + C$

(d) $\int x e^{\sin x^2} \cos x \, dx = \frac{1}{2} e^{\sin x^2} + C$

(e) $\int \frac{\sin x}{2 + \cos x} \, dx = -\ln|2 + \cos x| + C$

(f) $\int \frac{\sec^2(3x)}{1 + \tan(3x)} \, dx = \frac{1}{3} \ln|1 + \tan(3x)| + C$

(g) $\int \frac{\cos x - \sin x}{\sin x + \cos x} \, dx = \ln|\sin x + \cos x| + C$

(h) $\int \frac{e^{2x} + \cos(2x)}{e^{2x} + \sin(2x)} \, dx = \frac{1}{2} \ln|e^{2x} + \sin(2x)| + C$

(i) $\int 2 \sin x \cos^6 x \, dx$

(j) $\int 3 \tan^{10} x \sec^2 x \, dx$

$\int 3 \tan^{10} x \sec^2 x \, dx = \int 3(\tan x)^{10} \sec^2 x \, dx$

Step 1: Let $u = \tan x$

Step 2: $\frac{du}{dx} = \sec^2 x$ and $du = \sec^2 x \, dx$

Step 3: $\int 3(\tan x)^{10} \sec^2 x \, dx = \int 3u^{10} \, du$

Step 4: $= \frac{3u^{11}}{11} + C$

Step 5: $= \frac{3(\tan x)^{11}}{11} + C$

$\int 2 \sin x \cos^6 x \, dx = 2 \int \sin x (\cos x)^6 \, dx$

Step 1: Let $u = \cos x$

Step 2: $\frac{du}{dx} = -\sin x$ and $du = -\sin x \, dx \Rightarrow \frac{du}{-1} = \sin x \, dx$

Step 3: $\int 2 \sin x (\cos x)^6 \, dx = 2 \int (\cos x)^6 \sin x \, dx$

$= 2 \int u^6 \frac{du}{-1} = -2 \int u^6 \, du$

Step 4: $= -2 \times \frac{u^7}{7} + C = -\frac{2u^7}{7} + C$

Step 5: $= -\frac{2(\cos x)^7}{7} + C$